

PIGLET: Probabilistic Message Passing for Semi-supervised Link Sign Prediction

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Abstract—How can we accurately predict the signs of unseen links in partially observed signed graphs? Signed graphs are widely used to represent complex relationships in areas such as social and biological networks. Although prior methods enhance representation learning by extending Graph Neural Networks with social theories, they rely on the unrealistic assumption that all link signs are known. In practice, however, link signs are often only partially labeled due to the high cost or difficulty of obtaining ground-truth annotations. For example, in the Bitcoin transaction network, only a subset of interactions can be labeled as trusted or untrusted, while many remain unlabeled.

In this work, we propose PIGLET (PROBABILISTIC MESSAGE PASSING FOR SEMI-SUPERVISED LINK SIGN PREDICTION), a novel approach for accurate link sign prediction on signed graphs with partially observed sign labels. The main idea is to probabilistically interpret unlabeled links as both positive and negative based on a soft-labeling strategy, allowing information to be propagated through both types of edges. To mitigate the effect of uncertain predictions, we use confidence-based weights to downplay unreliable edges. PIGLET balances the importance of edges with relation-aware attention scores as well. PIGLET establishes a theoretical connection between the Expectation-Maximization (EM) algorithm and its iterative refinement of node embeddings and soft labels. Extensive experiments show that PIGLET consistently outperforms existing methods on real-world datasets under semi-supervised setting.

Index Terms—Semi-supervised learning, signed graphs, partially observed link signs, signed GNNs, link sign prediction

I. INTRODUCTION

How can we accurately predict signs of unseen links in signed graphs with partial labels? Signed graphs model both positive and negative relationships between nodes, and have been widely used to represent complex real-world systems, such as trust and distrust in social media [1], agreement and conflict in collaborative platforms [2], and activation and inhibition in biological networks [3]. Analyzing signed graphs has attracted attention from data mining communities [4] due to their diverse applications, including link sign prediction [1], recommender system [5], ranking [6], anomaly detection [7], etc. Predicting unknown signs of links is particularly important because it directly impacts the ability to understand and leverage relationship information within graphs.

Link sign prediction fundamentally depends on capturing the structural and relational properties of signed graphs. A promising way to achieve this is to learn effective node

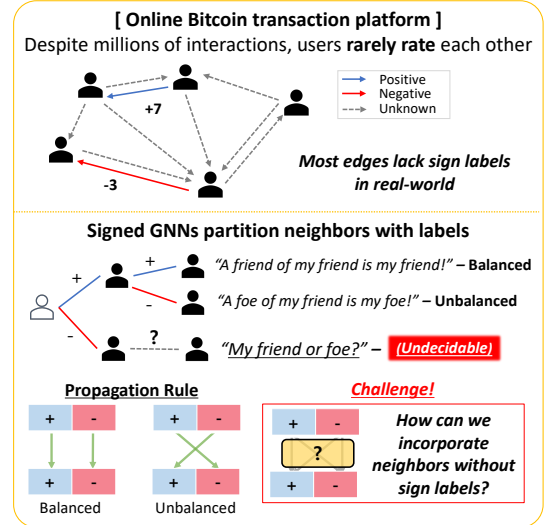


Fig. 1. Challenges in learning signed graphs with partially observed link signs. Explicit trust or distrust feedback is scarce in real-world graphs, leaving most links unlabeled. Existing SGNNs cannot handle such graphs effectively, as they assume that all link signs are available during training. This makes those social-theory-based methods struggle to aggregate information from unlabeled links, thereby degrading the quality of node embeddings.

representations that encode such graph characteristics. Among various approaches, Signed Graph Neural Networks (SGNNs) have been extensively adopted, extending traditional GNNs by considering link signs into message-passing process [8]. To better harness the unique characteristics of signed graphs, many researchers have proposed diverse SGNN methods that incorporate social theories [9], attention mechanisms [10], link directionality [11], bipartite properties [12], and more.

However, these previous methods [8]–[12] assume that signs of given links are fully observed when learning from signed graphs, even though this assumption may not hold in real-world scenarios [13]. For example, in peer-to-peer trading platforms, most trades occur without accompanying trust evaluations [14]. Similarly, on social networks, users frequently retweet or mention others without explicitly indicating endorsement or criticism [15], [16]. Such label scarcity is pervasive in recommender systems as well: platforms record countless implicit interactions such as clicks, views, and purchases, yet explicit sentiment labels, such as star ratings or thumbs-up/down, remain exceptionally rare [17], [18].

As a result, existing methods based on the assumption are not applicable to signed graphs with partially observed link signs, where rules underlying social theories or message propagation cannot be properly defined, as illustrated in Fig. 1. This limitation raises a fundamental question that motivates the need to account for the uncertainty introduced by unlabeled links: *how can we accurately predict the signs of unseen links by effectively leveraging both labeled and unlabeled relationships in partially-labeled signed graphs?*

To address this challenge, we propose PiGLET, an accurate link sign prediction method for partially-labeled signed graphs. Specifically, PiGLET infers probabilistic soft labels for unlabeled links using current node embeddings, enabling them to contribute to training. To suppress noisy signals from uncertain predictions, PiGLET performs confidence-weighted message passing, where each link’s influence is scaled by the reliability of its inferred label. Moreover, PiGLET employs relation-aware attention to capture relative relevance between target nodes and incoming neighbors across labeled and unlabeled edges. Since more accurate soft labels lead to improved node embeddings, PiGLET iteratively refines node embeddings and soft labels throughout training.

Our main contributions are summarized as follows:

- **Novel problem.** To the best of our knowledge, this is the first work to formalize the semi-supervised link sign prediction problem under realistic label sparsity.
- **Methodology.** We propose PiGLET, a novel semi-supervised learning method designed to accurately predict link signs by effectively leveraging both labeled and unlabeled links under partially observed signed graphs.
- **Theoretical analysis.** We establish a connection between our iterative refinement and the EM framework, offering theoretical justification for its gradual improvement of predictions and representations.
- **Experiments.** Extensive experiments under a semi-supervised learning setting show that PiGLET significantly outperforms baselines in link sign prediction.

The code and datasets for this study are publicly available at <https://github.com/snudatalab/PiGLEt>.

II. PRELIMINARIES AND RELATED WORK

We introduce key notations and formally define the problem addressed in this paper. Afterwards, we review prior studies on signed graph neural networks (SGNNs). Table I summarizes symbols frequently used in this paper.

A. Definitions

Partially observed signed graph. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denote a partially observed signed graph, where $\mathcal{V}$ is the set of nodes, and $\mathcal{E}$ is the set of edges, each of which is either labeled (signed) or unlabeled. Specifically, $\mathcal{E}$ is partitioned into a labeled subset $\mathcal{E}_L$ and an unlabeled subset $\mathcal{E}_U$, such that $\mathcal{E} = \mathcal{E}_L \cup \mathcal{E}_U$ and $\mathcal{E}_L \cap \mathcal{E}_U = \emptyset$. Let (u, v) denote a directed edge from node u to node v . If $(u, v) \in \mathcal{E}_L$, it is associated with a sign label $y_{uv} \in \{+1, -1\}$, indicating a positive or negative relation, respectively.

TABLE I
FREQUENTLY USED SYMBOLS.

Symbol	Description
$\mathcal{G} = (\mathcal{V}, \mathcal{E})$	Partially observed signed graph
$\mathcal{V}$	Set of nodes
$\mathcal{E}$	Set of edges
$\mathcal{E}_L$	Set of sign labeled edges
$\mathcal{E}_U$	Set of unlabeled edges
(u, v)	Directed edge from node u to node v
$\mathcal{N}_v^s$	Set of incoming neighbors connected to node v with label s , where $s \in \{+1, -1, ?\}$
$\mathcal{B}_v(l) \& \mathcal{U}_v(l)$	Set of balanced (or unbalanced) nodes connected to node v through paths of length l
y_{uv}	Ground-truth sign for $(u, v) \in \mathcal{E}_L$
$\hat{p}_{uv}$	Predicted probability of $(u, v) \in \mathcal{E}$ being positive
$\mathbf{h}_v^{\mathcal{B}(l)} \& \mathbf{h}_v^{\mathcal{U}(l)}$	Balanced / unbalanced embeddings of v at layer l

Signed and unlabeled in-neighbors. The set of incoming nodes connected to a destination node v with label s is defined as follows:

$$\mathcal{N}_v^s = \{u \in \mathcal{V} \mid y_{uv} = s\},$$

where $s \in \{+1, -1, ?\}$, and $y_{uv} = ?$ indicates that the edge’s label is unlabeled (or unknown).

Problem formulation. We formally define the problem addressed in this paper as follows.

Problem 1. (Semi-Supervised Link Sign Prediction)

- **Given:** a partially observed signed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ and initial node features $\{\mathbf{x}_v\}_{v \in \mathcal{V}}$
- **Predict:** the sign y_{uv} of an unseen edge (u, v) .

B. Structural Balance Theory

Most existing SGNNs [9], [19] leverage *structural balance theory* [20], [21] to more effectively encode node embeddings from signed graphs. The theory states that a triad (u, v, w) is *balanced* if the product of its three edge signs is positive, and *unbalanced* otherwise. Balanced motifs typically follow patterns such as “the friend of my friend is my friend,” while cases such as “the friend of my enemy is my friend” indicates unbalanced relationships. Motivated by the theory that balanced and unbalanced information convey different semantics, Derr et al. [9] defined sets of balanced and unbalanced nodes, extending the concept of balance to paths of arbitrary length, and aggregate information from each set differently.

Balanced and unbalanced nodes. For a destination node v , the set of balanced (or unbalanced) nodes consists of incoming nodes u such that the path from u to v contains an even (or odd) number of negative edges. Let $\mathcal{B}_v(l)$ and $\mathcal{U}_v(l)$ denote the sets of balanced and unbalanced nodes connected to node v via paths of length l . For $l = 1$, the sets are simply represented as $\mathcal{B}_v(1) = \mathcal{N}_v^+$ and $\mathcal{U}_v(1) = \mathcal{N}_v^-$. For $l > 1$, the balanced set is defined as:

$$\mathcal{B}_v(l+1) = \{w \mid u \in \mathcal{B}_v(l), w \in \mathcal{N}_u^+\} \cup \{w \mid u \in \mathcal{U}_v(l), w \in \mathcal{N}_u^-\},$$

while the unbalanced set is defined as:

$$\mathcal{U}_v(l+1) = \{w \mid u \in \mathcal{B}_v(l), w \in \mathcal{N}_u^-\} \cup \{w \mid u \in \mathcal{U}_v(l), w \in \mathcal{N}_u^+\}.$$

C. Signed Graph Neural Networks

Graph Neural Networks (GNNs) have been extensively studied due to their effectiveness in modeling relational data across diverse domains, leading to a wide range of research directions [22]–[28]. Signed Graph Neural Networks (SGNNs) extend traditional GNNs by explicitly considering the semantics of positive and negative relationships during message passing [8]. In particular, assuming that the edge labels in a given signed graph are fully observed, most SGNNs [9], [10], [12], [19] incorporate social theories (i.e., balance or status theory [1], [20], [21]) to more effectively capture the nature of signed relationships. As a pioneering work, SGCN [9] extends Graph Convolutional Network [29] by incorporating balance theory, and aggregates information separately from balanced and unbalanced neighbors using distinct weights. SiGAT [10] generalizes Graph Attention Network [30] to signed graphs. SDGNN [11] incorporates both edge directions and signs into attention-based message passing. SNEA [19] refines edge-level representations through sign-aware neighbor selection and gated aggregation. Recent SGNN studies have explored strategies such as graph diffusion [31], magnetic Laplacians [32], [33], clustering [34], and disentanglement [35].

However, the existing methods cannot be applied to partially observed signed graphs in Problem 1, as they assume that all edge signs are available in the input graph under a fully supervised learning setting. Although learning under label scarcity has been extensively explored in various fields [36]–[39], research in the context of signed graphs remains under-explored despite its importance. The presence of unlabeled edges makes it difficult to identify potentially balanced or unbalanced nodes, introducing uncertainty into the message passing process, as shown in Fig. 1. In contrast, our proposed PIGLET effectively addresses this uncertainty within a semi-supervised learning setting, resulting in more expressive node embeddings for link sign prediction, as validated in Section IV.

III. PROPOSED METHOD

We propose PIGLET, a novel approach to accurately predict signs of unseen edges from partially observed signed graphs under semi-supervised learning setting. We summarize the key technical challenges and our solutions as follows:

- C1. How can we effectively utilize unlabeled edges when labeled data are limited?** As described in Problem 1, most edges are unlabeled, limiting supervision. To address this, we estimate probabilistic soft labels for unlabeled edges, where these soft labels represent the likelihood of each edge being positive or negative, enabling each edge to simultaneously propagate information as both positive and negative (Section III-A).
- C2. How can we handle uncertainty in soft-labeled edges during aggregation?** Naively treating soft-labeled edges as labeled ones can introduce noise and unreliability into message aggregation. For this, we propose a confidence-weighted aggregation that scales each edge’s contribution by the confidence of its inferred label, amplifying reliable signals while suppressing uncertain ones (Section III-B).

Algorithm 1: Overall Process of PIGLET

Input: Partially observed signed graph $G = (\mathcal{V}, \mathcal{E}_L, \mathcal{E}_U)$; number L of layers; number T of epochs; initial node embeddings $\{\mathbf{x}_v\}$; model parameters θ ; learning rate η

Output: Final node embeddings $\{\mathbf{h}_v^{(T)}\}$

- 1 Initialize the hidden embedding $\mathbf{h}_v^{(0)} \leftarrow \mathbf{x}_v, \forall v \in \mathcal{V}$
- 2 **for** $t = 1$ to T **do**
- 3 **for** $(u, v) \in \mathcal{E}_U$ **do**
- 4 Predict the soft label p_{uv}^+ with $\mathbf{h}_u^{(t-1)}$ & $\mathbf{h}_v^{(t-1)}$ $\triangleright$ Eq. (1)
- 5 Compute the confidence score c_{uv} from p_{uv}^+ $\triangleright$ Eq. (4)
- 6 Initialize embeddings $\mathbf{h}_v^{B(0)}$ & $\mathbf{h}_v^{U(0)} \leftarrow \mathbf{h}_v^{(t-1)}, \forall v \in \mathcal{V}$
- 7 **for** $l = 1$ to L **do**
- 8 **for** $v \in \mathcal{V}$ **do**
- 9 Compute the relation-aware attention scores $\alpha_{uv}^{S \rightarrow T(l)}$ from neighbors u to v for $S, T \in \{\mathcal{B}, \mathcal{U}\}$ $\triangleright$ Eq. (5)
- 10 Compute $\mathbf{h}_v^{B(l)}$ & $\mathbf{h}_v^{U(l)}$ through aggregations with confidences $\{c_{uv}\}$ and attentions $\{\alpha_{uv}^{S \rightarrow T(l)}\}$ $\triangleright$ Eqs. (6) & (7)
- 11 Compute $\mathbf{h}_v^{(t)}$ by merging $\mathbf{h}_v^{B(L)}$ & $\mathbf{h}_v^{U(L)}$, $\forall v \in \mathcal{V}$ $\triangleright$ Eq. (8)
- 12 Compute $\mathcal{L}_{\text{total}}$ and update $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}_{\text{total}}$ $\triangleright$ Eq. (9)
- 13 **return** $\{\mathbf{h}_v^{(T)}\}$

C3. How can we design the loss function to effectively utilize unlabeled edges? We construct a soft label-based self-training loss for unlabeled edges, guided by a supervised loss of labeled edges. Additionally, we integrate a structural regularization loss to enforce embeddings of positively connected nodes to be closer, and those of negatively connected nodes to be farther apart (Section III-C).

Fig. 2 illustrates the overall process of PIGLET. PIGLET computes node embeddings from the graph with only labeled edges, forming an initial basis for inference. These embeddings then inform probabilistic predictions for unlabeled edges, producing soft labels that quantify the likelihood of each edge being positive or negative. Soft labels and embeddings are progressively refined through epochs, achieving more accurate link sign prediction and expressive node representations. We present the high-level pseudocode for our iterative refinement process in Algorithm 1.

A. Iterative Learning with Soft Labels

In signed graphs, positive and negative edges have fundamentally different semantics: a *friend* typically reinforces a node’s beliefs, whereas an *enemy* tends to contradict or weaken them. This contrasting influence is modeled by separately aggregating respective embeddings along balanced (positive) and unbalanced (negative) paths.

However, unlabeled edges complicate this scenario, as their uncertain polarity obscures the correct aggregation path. Discarding unlabeled edges leads to loss of valuable relational information, while naively assigning hard labels to uncertain edges risks propagating unreliable information across the network. To address such limitation, we interpret every unlabeled neighbor as both a *potential* friend and enemy. PIGLET takes a probabilistic view, allowing the model to express potential sign of each unlabeled edge in a principled manner.

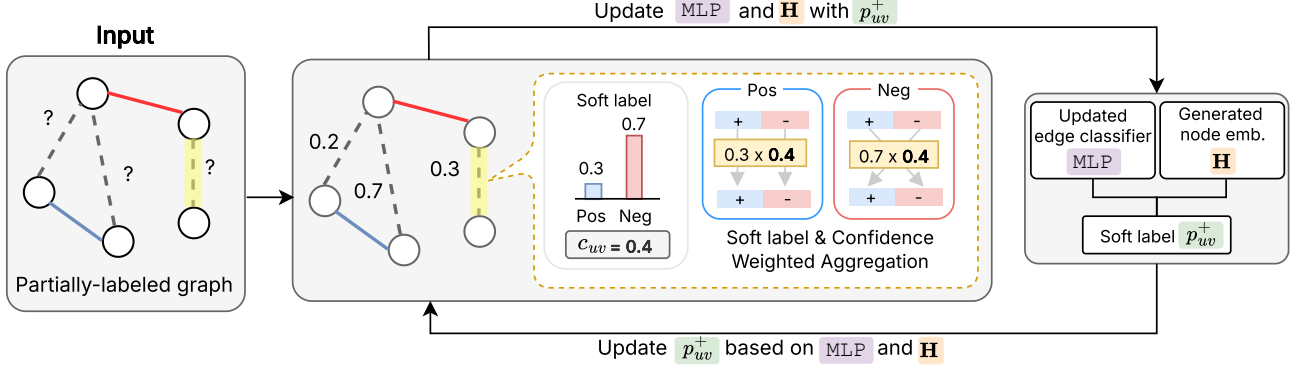


Fig. 2. Overview of PiGLET. Given a partially observed signed graph $\mathcal{G}$, PiGLET first assigns soft labels p_{uv}^+ to unlabeled edges using node embeddings $\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_{|\mathcal{V}|}]$ and an edge classifier MLP. These soft-labels are then integrated into relation-aware message passing, weighted by their corresponding confidence scores. The resulting node embeddings are used iteratively to update the edge classifier and refine soft-labels, progressively enhancing both the embedding quality and the accuracy of inferred edge signs.

Probabilistic soft labeling. For each unlabeled neighbor $u \in \mathcal{N}_v^?$, we iteratively refine the soft label of the edge (u, v) during training. At every epoch t , we obtain the final node embedding $\mathbf{h}_v^{(t)}$ for node $v \in \mathcal{V}$ (details in Section III-B), which is then used to update the soft labels for the next epoch. The probabilistic soft label p_{uv}^+ of an unlabeled edge (u, v) indicates the likelihood of treating it as either positive or negative at epoch t , computed as follows:

$$p_{uv}^+ = \sigma \left(\text{MLP} \left([\mathbf{h}_u^{(t-1)} \parallel \mathbf{h}_v^{(t-1)}] \right) \right), \quad (1)$$

where $\mathbf{h}_u^{(t-1)}$ and $\mathbf{h}_v^{(t-1)}$ denote the embeddings of nodes u and v obtained at epoch $t-1$, respectively. We use multi-layer perceptron (MLP) with one hidden layer and $\sigma(\cdot)$ is the sigmoid activation function. Note that at the initial stage ($t=1$), we set p_{uv}^+ to 0.5, representing an equal likelihood of being either positive or negative. For $u \in \mathcal{N}_v^+$, we treat u as an in-neighbor positively connected to v with probability p_{uv}^+ , and negatively connected with probability $1 - p_{uv}^+$.

B. Reflecting Uncertainty in Message Passing

Existing SGNNs are limited in handling unlabeled edges as they rely solely on structural balance theory, categorizing neighbors strictly into balanced and unbalanced sets according to their relational paths (see Section II-B). To address this, PiGLET interprets each unlabeled edge as potentially both positive and negative, allowing it to simultaneously participate in both balanced and unbalanced paths during message passing. This inclusive two-way sign interpretation enables information flow even without ground-truth signs, allowing the model to effectively utilize unlabeled edges during training.

We maintain two relation-specific embeddings for each node $v \in \mathcal{V}$: $\mathbf{h}_v^{\mathcal{B}(l)}$ for balanced (positive) and $\mathbf{h}_v^{\mathcal{U}(l)}$ for unbalanced (negative) relationships, each after l layers of relation-specific message passing. As positive neighbors $u \in \mathcal{N}_v^+$ reinforce the target node, their embeddings follow same-channel propagation paths ($\mathcal{B} \rightarrow \mathcal{B}$, $\mathcal{U} \rightarrow \mathcal{U}$). Conversely, contrasting influence of negative neighbors $w \in \mathcal{N}_v^-$ are formulated via cross-channel propagation paths ($\mathcal{U} \rightarrow \mathcal{B}$, $\mathcal{B} \rightarrow \mathcal{U}$).

Formally, $\mathbf{h}_v^{\mathcal{B}(l)}$ and $\mathbf{h}_v^{\mathcal{U}(l)}$ of the target node v are aggregated as follows:

$$\begin{aligned} \mathbf{h}_v^{\mathcal{B}(l)} = & \sum_{u \in \mathcal{N}_v^+} \mathbf{h}_u^{\mathcal{B} \rightarrow \mathcal{B}(l)} + \sum_{w \in \mathcal{N}_v^-} \mathbf{h}_w^{\mathcal{U} \rightarrow \mathcal{B}(l)} \\ & + \sum_{k \in \mathcal{N}_v^?} (p_{kv}^+ \mathbf{h}_k^{\mathcal{B} \rightarrow \mathcal{B}(l)} + (1 - p_{kv}^+) \mathbf{h}_k^{\mathcal{U} \rightarrow \mathcal{B}(l)}), \end{aligned} \quad (2)$$

$$\begin{aligned} \mathbf{h}_v^{\mathcal{U}(l)} = & \sum_{u \in \mathcal{N}_v^+} \mathbf{h}_u^{\mathcal{U} \rightarrow \mathcal{U}(l)} + \sum_{w \in \mathcal{N}_v^-} \mathbf{h}_w^{\mathcal{B} \rightarrow \mathcal{U}(l)} \\ & + \sum_{k \in \mathcal{N}_v^?} (p_{kv}^+ \mathbf{h}_k^{\mathcal{U} \rightarrow \mathcal{U}(l)} + (1 - p_{kv}^+) \mathbf{h}_k^{\mathcal{B} \rightarrow \mathcal{U}(l)}), \end{aligned} \quad (3)$$

where p_{kv}^+ and $1 - p_{kv}^+$ are the soft labels computed by Eq. (1). Note that there are four types of embeddings $\mathbf{h}_u^{\mathcal{B} \rightarrow \mathcal{B}(l)}$, $\mathbf{h}_u^{\mathcal{U} \rightarrow \mathcal{B}(l)}$, $\mathbf{h}_u^{\mathcal{B} \rightarrow \mathcal{U}(l)}$, and $\mathbf{h}_u^{\mathcal{U} \rightarrow \mathcal{U}(l)}$ for each node u to reflect distinct relational semantics. For example, $\mathbf{h}_u^{\mathcal{B} \rightarrow \mathcal{U}(l)}$ denotes the balanced embedding of a source node u used to update the unbalanced embedding of a target node (e.g., v in Eq. (3)). These embeddings are computed as follows:

$$\begin{aligned} \mathbf{h}_u^{\mathcal{B} \rightarrow \mathcal{B}(l)} & \leftarrow \mathbf{W}^{\mathcal{B}(l)} \mathbf{h}_u^{\mathcal{B}(l-1)} & (\text{Balanced to Balanced}), \\ \mathbf{h}_u^{\mathcal{U} \rightarrow \mathcal{U}(l)} & \leftarrow \mathbf{W}^{\mathcal{U}(l)} \mathbf{h}_u^{\mathcal{U}(l-1)} & (\text{Unbalanced to Unbalanced}), \\ \mathbf{h}_w^{\mathcal{U} \rightarrow \mathcal{B}(l)} & \leftarrow \mathbf{W}^{\mathcal{B}(l)} \mathbf{h}_w^{\mathcal{U}(l-1)} & (\text{Unbalanced to Balanced}), \\ \mathbf{h}_w^{\mathcal{B} \rightarrow \mathcal{U}(l)} & \leftarrow \mathbf{W}^{\mathcal{U}(l)} \mathbf{h}_w^{\mathcal{B}(l-1)} & (\text{Balanced to Unbalanced}), \end{aligned}$$

where $\mathbf{W}^{\mathcal{B}(l)}$ and $\mathbf{W}^{\mathcal{U}(l)}$ are learnable weight matrices capturing different relational semantics inherent in balanced and unbalanced relationships. For simplicity, bias terms are omitted in the equations, although they are used in our implementation.

However, there are two subproblems with directly applying p_{uv}^+ and $1 - p_{uv}^+$ as weights for message passing. In the following, we describe the problems and propose our solutions.

Subproblem 1. The first challenge lies in the difficulty of suppressing the influence of uncertain edges. For instance, when an edge has predicted probabilities $[p_{uv}^+, 1 - p_{uv}^+]$ close to $[0.5, 0.5]$, it indicates high uncertainty, with no clear preference for either a positive or negative sign. Allowing such an uncertain edge to contribute equally as a confidently predicted one (e.g., $[0.9, 0.1]$) is undesirable, as it may introduce noisy signals that hinder the learning of meaningful representations.

Solution 1: Confidence-based weights. We propose a confidence-based weight c_{uv} to diminish the influence of edges with uncertain predictions. Edges with probabilities near 0 or 1 carry greater weight, while those around 0.5 have limited influence. We define c_{uv} as follows:

$$c_{uv} = 2 \cdot |p_{uv}^+ - 0.5| \in [0, 1]. \quad (4)$$

c_{uv} approaches 1 as p_{uv}^+ becomes more certain (i.e., closer to 0 or 1), and is used as an additional weight during message propagation to modulate the influence of unlabeled edges.

Subproblem 2. The second challenge is that the aggregation functions in Eqs. (2) and (3) fail to holistically capture the importance of labeled and unlabeled neighbors together. While unlabeled edges provide implicit and uncertain relationships, labeled edges carry explicit and highly reliable relational information. Thus, contributions of both labeled and unlabeled neighbors should also be adaptively weighted.

Solution 2: Relation-aware attentions. We propose a relation-aware attention mechanism that adaptively weights each neighbor according to its feature similarity with the target node. With this approach, the model is enabled to integratively capture relative importance among labeled and unlabeled neighbors regardless of label availability. We aim to model diverse attention signals between balanced and unbalanced semantics to more precisely capture the relationship of an edge (u, v) . Thus, we model the attention probability $\alpha_{uv}^{S \rightarrow T(l)}$ to represent the importance of neighbor $u \in \mathcal{N}_v$ from relation $\mathcal{S}$ when propagating messages to target node v under relation $\mathcal{T}$.

Since each combination of relation types $\mathcal{S}, \mathcal{T} \in \{\mathcal{B}, \mathcal{U}\}$ specifies a unique propagation rule, the corresponding attention probability is computed as follows:

$$\alpha_{uv}^{S \rightarrow T(l)} = \frac{\exp(e_{uv}^{S \rightarrow T(l)})}{\sum_{\mathcal{R} \in \{\mathcal{B}, \mathcal{U}\}} \sum_{k \in \mathcal{N}_v} \exp(e_{kv}^{\mathcal{R} \rightarrow T(l)})}, \quad (5)$$

where $\mathcal{B}$ and $\mathcal{U}$ are balanced and unbalanced relations, respectively. A neighbor u with features similar to those of the target node v receives a higher $\alpha_{uv}^{S \rightarrow T(l)}$. The attention probability $\alpha_{uv}^{S \rightarrow T(l)}$ is used as an additional weight of an edge (u, v) during message passing in Eqs. (6) and (7). We first compute the attention coefficient for every neighbor. Attention coefficient $e_{uv}^{S \rightarrow T(l)}$ in Eq. (5) is computed by applying an attention mechanism $\text{att}(\cdot)$ as follows:

$$e_{uv}^{S \rightarrow T(l)} = \text{att}(\mathbf{h}_u^{S \rightarrow T(l)}, \mathbf{h}_v^{S \rightarrow T(l)}).$$

To implement $\text{att}(\cdot)$, we concatenate the embeddings of u and v w.r.t. the relations, apply a linear transformation, and then a hyperbolic tangent function.

Aggregating hidden node embeddings. We extend the initial aggregation rules in Eqs. (2) and (3) by integrating the confidence-based weights c_{uv} in Eq. (4) and the relation-aware attentions $\alpha_{uv}^{S \rightarrow T(l)}$ in Eq. (5). For unlabeled edges, an edge with high c_{uv} propagates information similarly to labeled edges, thereby making active contribution based on its relation-aware importance $\alpha_{uv}^{S \rightarrow T(l)}$. Conversely, even if an edge has a high $\alpha_{uv}^{S \rightarrow T(l)}$, its influence is diminished by a low c_{uv} resulting from an uncertain prediction, thus limiting its overall contribution.

For the target node v , we obtain the balanced embedding $\mathbf{h}_v^{\mathcal{B}(l)}$ and unbalanced one $\mathbf{h}_v^{\mathcal{U}(l)}$ at layer l as follows:

$$\begin{aligned} \mathbf{h}_v^{\mathcal{B}(l)} = & \sum_{u \in \mathcal{N}_v^+} \alpha_{uv}^{\mathcal{B} \rightarrow \mathcal{B}(l)} \mathbf{h}_u^{\mathcal{B} \rightarrow \mathcal{B}(l)} + \sum_{w \in \mathcal{N}_v^-} \alpha_{vw}^{\mathcal{U} \rightarrow \mathcal{B}(l)} \mathbf{h}_w^{\mathcal{U} \rightarrow \mathcal{B}(l)} \\ & + \sum_{k \in \mathcal{N}_v^?} \left(c_{kv} \cdot p_{kv}^+ \cdot \alpha_{kv}^{\mathcal{B} \rightarrow \mathcal{B}(l)} \mathbf{h}_k^{\mathcal{B} \rightarrow \mathcal{B}(l)} \right. \\ & \left. + c_{kv} \cdot (1 - p_{kv}^+) \cdot \alpha_{kv}^{\mathcal{U} \rightarrow \mathcal{B}(l)} \mathbf{h}_k^{\mathcal{U} \rightarrow \mathcal{B}(l)} \right), \end{aligned} \quad (6)$$

$$\begin{aligned} \mathbf{h}_v^{\mathcal{U}(l)} = & \sum_{u \in \mathcal{N}_v^+} \alpha_{uv}^{\mathcal{U} \rightarrow \mathcal{U}(l)} \mathbf{h}_u^{\mathcal{U} \rightarrow \mathcal{U}(l)} + \sum_{w \in \mathcal{N}_v^-} \alpha_{vw}^{\mathcal{B} \rightarrow \mathcal{U}(l)} \mathbf{h}_w^{\mathcal{B} \rightarrow \mathcal{U}(l)} \\ & + \sum_{k \in \mathcal{N}_v^?} \left(c_{kv} \cdot p_{kv}^+ \cdot \alpha_{kv}^{\mathcal{U} \rightarrow \mathcal{U}(l)} \mathbf{h}_k^{\mathcal{U} \rightarrow \mathcal{U}(l)} \right. \\ & \left. + c_{kv} \cdot (1 - p_{kv}^+) \cdot \alpha_{kv}^{\mathcal{B} \rightarrow \mathcal{U}(l)} \mathbf{h}_k^{\mathcal{B} \rightarrow \mathcal{U}(l)} \right). \end{aligned} \quad (7)$$

Each unlabeled edge's contribution is modulated by its confidence c_{uv} , ensuring edges with near-certain labels (p_{uv}^+ close to 0 or 1, high c_{uv}) strongly inform the embedding, while uncertain edges ($p_{uv}^+ \approx 0.5$, low c_{uv}) are down-weighted.

After passing through the L -th layer, the final embedding $\mathbf{h}_v^{(t)}$ of node v at epoch t is computed as follows:

$$\mathbf{h}_v^{(t)} = \tanh \left(\mathbf{W}_M [\mathbf{h}_v^{\mathcal{B}(L)} \parallel \mathbf{h}_v^{\mathcal{U}(L)}] \right), \quad (8)$$

where $\mathbf{W}_M$ is a learnable weight matrix and $\parallel$ denotes concatenation. This final embedding then serves as the input for updating probabilistic soft labels p_{uv}^+ and $1 - p_{uv}^+$ in the next epoch, thereby progressively enhancing prediction accuracy and embedding quality during training.

C. Objective Functions

We aim to effectively 1) leverage a limited number of labeled edges, 2) exploit abundant unlabeled edges, and 3) preserve the relational structures inherent in signed graphs for better representation learning. For this purpose, we jointly optimize 1) a supervised binary cross-entropy (BCE) loss $\mathcal{L}_L$ on labeled edges, 2) a self-training BCE loss $\mathcal{L}_U$ on unlabeled edges, and 3) a structure-preserving hinge loss $\mathcal{L}_S$ based on balance theory [19].

BCE losses. The supervised BCE loss $\mathcal{L}_L$ for labeled edges $\mathcal{E}_L$ is defined as follows:

$$\mathcal{L}_L = - \sum_{(u,v) \in \mathcal{E}_L} (y_{uv} \log \hat{p}_{uv} + (1 - y_{uv}) \log(1 - \hat{p}_{uv})),$$

where y_{uv} is the ground-truth sign of (u, v) , and $\hat{p}_{uv}$ is the predicted probability of being positive.

The self-training loss $\mathcal{L}_U$ on unlabeled edges $\mathcal{E}_U$ is formulated as follows:

$$\mathcal{L}_U = - \sum_{(u,v) \in \mathcal{E}_U} (p_{uv}^+ \log \hat{p}_{uv} + (1 - p_{uv}^+) \log(1 - \hat{p}_{uv})),$$

where p_{uv}^+ is the soft label of (u, v) obtained by Eq. (1), and $\hat{p}_{uv}$ is the predicted probability.

Structure-preserving loss. We also optimize the structure-preserving loss $\mathcal{L}_S$, inspired by structural balance theory, to inject structural information into the representations [19].

Specifically, the theory encourages embeddings of positively connected nodes to be closer than those of unrelated pairs, and embeddings of negatively connected nodes to be more

distant than those of pairs without negative connections. For this purpose, we define the following sets:

$$T^+ = \{(u, v, w) \mid u \in \mathcal{N}_v^+, w \notin \mathcal{N}_v^+\},$$

$$T^- = \{(k, v, w) \mid k \in \mathcal{N}_v^-, w \notin \mathcal{N}_v^-\},$$

where each triplet in T^+ consists of a positively connected node pair (u, v) and a node w not positively connected to v . Similarly, each triplet in T^- consists of a negatively connected node pair (k, v) and a node w not negatively connected to v . Note that the number of triples in T^s is equal to the number of labeled edges with sign s . We penalize violations of these triplet margin constraints as follows:

$$\mathcal{L}_S = \frac{1}{|T^+|} \sum_{(u,v,w) \in T^+} \max(0, \|\mathbf{h}_u - \mathbf{h}_v\|^2 - \|\mathbf{h}_v - \mathbf{h}_w\|^2)$$

$$+ \frac{1}{|T^-|} \sum_{(k,v,w) \in T^-} \max(0, \|\mathbf{h}_v - \mathbf{h}_w\|^2 - \|\mathbf{h}_k - \mathbf{h}_v\|^2).$$

Total loss. The total loss $\mathcal{L}_{\text{total}}$ is defined as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_L + \lambda_U \mathcal{L}_U + \lambda_S \mathcal{L}_S, \quad (9)$$

where λ_U and λ_S control the weights of their corresponding losses. As training progresses, $\mathcal{L}_S$ enforces balance-theoretic consistency, while $\mathcal{L}_L$ and $\mathcal{L}_U$ guide edge sign predictions.

D. Theoretical Analysis

We analyze our proposed PiGLET within the framework of the Expectation-Maximization (EM) algorithm, motivated by [40]. The EM algorithm iteratively estimates model parameters by maximizing the expected log-likelihood of data under partial observation. Given the observed target variables $\mathbf{y}$, the input node features $\mathbf{X}$, and the current estimate $\theta^{(t)}$ of model parameters, the EM algorithm introduces latent variables $\mathbf{z}$ for the unobserved data. The goal is to iteratively maximize the expected complete-data log-likelihood $Q(\theta \mid \theta^{(t)})$ to obtain the updated estimate θ of parameters, formulated as:

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z} \mid \mathbf{y}, \theta^{(t)})} [\log p(\mathbf{y}, \mathbf{z} \mid \theta)], \quad (10)$$

where we omit the explicit dependence on $\mathbf{X}$ for brevity.

The posterior $p(\mathbf{z} \mid \mathbf{y}, \theta^{(t)})$ captures dependencies among unlabeled edge signs induced by the graph structure. Since each edge $(u, v) \in \mathcal{E}_U$ can be either positive or negative, its exact computation requires summing over $2^{|\mathcal{E}_U|}$ assignments, which is computationally infeasible. Thus, we approximate the posterior by treating each unlabeled edge independently with a classifier $f_{\theta^{(t)}}$ parameterized by $\theta^{(t)}$ as follows:

$$p(\mathbf{z} \mid \mathbf{y}, \theta^{(t)}) \approx \prod_{(u,v) \in \mathcal{E}_U} f_{\theta^{(t)}}(z_{uv}), \quad (11)$$

where $f_{\theta^{(t)}}(z_{uv})$ returns the probability of assigning latent sign z_{uv} to an unlabeled edge $(u, v) \in \mathcal{E}_U$, corresponding to the following soft labels:

$$f_{\theta^{(t)}}(z_{uv}) = \begin{cases} p_{uv}^+, & \text{if } z_{uv} = +1, \\ 1 - p_{uv}^+, & \text{if } z_{uv} = -1. \end{cases}$$

The joint distribution over the observed labels $\mathbf{y}$ for edges in $\mathcal{E}_L$ and latent labels $\mathbf{z}$ for edges in $\mathcal{E}_U$, conditioned on the parameters θ , is similarly approximated using f_{θ} :

$$p(\mathbf{y}, \mathbf{z} \mid \theta) \approx \prod_{(u,v) \in \mathcal{E}_L} f_{\theta}(y_{uv}) \prod_{(u,v) \in \mathcal{E}_U} f_{\theta}(z_{uv}), \quad (12)$$

where the function f_{θ} for edge (u, v) is defined as:

$$f_{\theta}(\ell_{uv}) = \begin{cases} \hat{p}_{uv}, & \ell_{uv} = +1, \\ 1 - \hat{p}_{uv}, & \ell_{uv} = -1, \end{cases}$$

where $\hat{p}_{uv} = \sigma(\text{MLP}([\mathbf{h}_u^{(t)} \parallel \mathbf{h}_v^{(t)}]))$.

Theorem 1. Given the approximations in Eqs. (11) and (12), the EM objective $Q(\theta \mid \theta^{(t)})$ reduces to $\mathcal{L}_{\text{total}}$ with $\lambda_U = 1$ and $\lambda_S = 0$ in Eq. (9), represented as follows:

$$Q(\theta \mid \theta^{(t)}) \approx -(\mathcal{L}_L + \mathcal{L}_U).$$

Proof. By the definition of expectation, the EM objective $Q(\theta \mid \theta^{(t)})$ of Eq. (10) is expressed as:

$$Q(\theta \mid \theta^{(t)}) = \sum_{\mathbf{z}} p(\mathbf{z} \mid \mathbf{y}, \theta^{(t)}) \log p(\mathbf{y}, \mathbf{z} \mid \theta).$$

It is then approximated using Eqs. (11) and (12) as follows:

$$Q(\theta \mid \theta^{(t)}) \approx \sum_{\mathbf{z}} \prod_{(u,v) \in \mathcal{E}_U} f_{\theta^{(t)}}(z_{uv}) \log \left(\prod_{(u,v) \in \mathcal{E}_L} f_{\theta}(y_{uv}) \prod_{(u,v) \in \mathcal{E}_U} f_{\theta}(z_{uv}) \right)$$

$$= \sum_{(u,v) \in \mathcal{E}_L} \log(f_{\theta}(y_{uv})) + \underbrace{\sum_{\mathbf{z}} \prod_{(u,v) \in \mathcal{E}_U} f_{\theta^{(t)}}(z_{uv}) \sum_{(u,v) \in \mathcal{E}_U} \log(f_{\theta}(z_{uv}))}_{\mathcal{F}(\theta \mid \theta^{(t)})},$$

where the term $\sum_{\mathbf{z}} \prod_{(u,v) \in \mathcal{E}_U} f_{\theta^{(t)}}(z_{uv})$ in the first part marginalizes to 1, as it sums over all possible sign assignments for unlabeled edges and is independent of the labeled data. Suppose $\mathcal{E}_U = \{e_1, \dots, e_M\}$ and $\mathcal{C} = \{-1, +1\}$. Then, the second part $\mathcal{F}(\theta \mid \theta^{(t)})$ is represented as:

$$\mathcal{F}(\theta \mid \theta^{(t)}) = \sum_{\mathbf{z}} \prod_{(u,v) \in \mathcal{E}_U} f_{\theta^{(t)}}(z_{uv}) \sum_{(u,v) \in \mathcal{E}_U} \log(f_{\theta}(z_{uv}))$$

$$= \sum_{z_{e_1} \in \mathcal{C}} \cdots \sum_{z_{e_M} \in \mathcal{C}} [f_{\theta^{(t)}}(z_{e_1}) \cdots f_{\theta^{(t)}}(z_{e_M})]$$

$$\times [\log f_{\theta}(z_{e_1}) + \cdots + \log f_{\theta}(z_{e_M})]$$

$$= \sum_{i=1}^M \sum_{\mathbf{z}} f_{\theta^{(t)}}(z_{e_1}) \cdots f_{\theta^{(t)}}(z_{e_M}) \log f_{\theta}(z_{e_i}).$$

By separating the log terms independently,

$$\mathcal{F}(\theta \mid \theta^{(t)}) = \sum_{i=1}^M \sum_{z_{e_i} \in \mathcal{C}} f_{\theta^{(t)}}(z_{e_i}) \log f_{\theta}(z_{e_i}) \times \underbrace{\prod_{j \neq i} \sum_{z_{e_j} \in \mathcal{C}} f_{\theta^{(t)}}(z_{e_j})}_{=1}$$

$$= \sum_{(u,v) \in \mathcal{E}_U} \sum_{z_{uv} \in \mathcal{C}} f_{\theta^{(t)}}(z_{uv}) \log f_{\theta}(z_{uv}).$$

Thus, $Q(\theta \mid \theta^{(t)})$ is represented as follows:

$$Q(\theta \mid \theta^{(t)}) \approx \sum_{(u,v) \in \mathcal{E}_L} [\log(f_{\theta}(y_{uv}))] + \mathcal{F}(\theta \mid \theta^{(t)})$$

$$= \sum_{(u,v) \in \mathcal{E}_L} \log(f_{\theta}(y_{uv})) + \sum_{(u,v) \in \mathcal{E}_U} \sum_{z_{uv} \in \mathcal{C}} f_{\theta^{(t)}}(z_{uv}) \log(f_{\theta}(z_{uv}))$$

$$= \sum_{(u,v) \in \mathcal{E}_L} [y_{uv} \log(\hat{p}_{uv}) + (1 - y_{uv}) \log(1 - \hat{p}_{uv})]$$

$$+ \sum_{(u,v) \in \mathcal{E}_U} [p_{uv}^+ \log(\hat{p}_{uv}) + (1 - p_{uv}^+) \log(1 - \hat{p}_{uv})]$$

$$= -(\mathcal{L}_L + \mathcal{L}_U),$$

where $y_{uv} \in \mathcal{C}$ and the result is equal to $-\mathcal{L}_{\text{total}}$ with $\lambda_U = 1$ and $\lambda_S = 0$. $\square$

Thus, if we assume that the gradient update performed at each epoch sufficiently optimizes the model parameters, the iterative process of updating soft labels and subsequently learning from these updated labels closely resembles the EM algorithm, which iteratively updates the latent-label distribution and model parameters. This equivalence clarifies the theoretical interpretation of each training iteration in PiGLET as a special case of the EM framework.

Time complexity. Theorem 2 shows that PiGLET scales linearly with the number $|\mathcal{E}|$ of edges, assuming that d and L are constant, and $|\mathcal{V}| < |\mathcal{E}|$ as it is typical in real-world sparse graphs.

Theorem 2. *The time complexity of Algorithm 1 of PiGLET is $O(d^2|\mathcal{E}_U| + L(d^2|\mathcal{V}| + d|\mathcal{E}|))$ per epoch, where d is the embedding dimension and L is the number of layers.*

Proof. PiGLET proceeds in two main steps: (1) computing soft labels for unlabeled edges and (2) performing weighted propagation across all edges. Computing the soft labels p_{uv}^+ and confidence scores c_{uv} for unlabeled edges results in $O(d^2|\mathcal{E}_U|)$. At each layer, feature transformation for all nodes takes $O(d^2|\mathcal{V}|)$. Attention score computation and normalization across four relation types over all edges incur $O(d|\mathcal{E}|)$. The message passing step takes $O(d|\mathcal{N}_v|)$ for each node v , resulting in total $O(d|\mathcal{E}|)$ across all nodes. The final node embedding update requires $O(d^2|\mathcal{V}|)$, and computing the losses takes $O(|\mathcal{E}|)$. Thus, the total time complexity is $O(d^2|\mathcal{E}_U| + L(d^2|\mathcal{V}| + d|\mathcal{E}|))$. $\square$

IV. EXPERIMENTS

We conduct experiments to answer the following questions:

- Q1. Performance comparison (Section IV-B).** How well does PiGLET perform in predicting link signs compared to baseline methods under a semi-supervised setting?
- Q2. Ablation study (Section IV-C).** How does each component of PiGLET contribute to its overall performance?
- Q3. Confidence analysis (Section IV-D).** How do the confidence scores measured by PiGLET change throughout training, and do they become more reliable over time?
- Q4. Scalability (Section IV-E).** Does PiGLET exhibit linear complexity with respect to the number of edges?
- Q5. Effect of hyperparameters (Section IV-F).** How do the hyperparameters of PiGLET impact its performance?

A. Experimental Settings

Datasets. We use five real-world signed graphs [41], whose statistics are summarized in Table II. $\mathcal{V}$ and $\mathcal{E}$ denote the number of nodes and edges, respectively; $|\mathcal{E}^s|$ denotes the number of edges with sign $s \in \{+1, -1\}$; and $\rho(+) = |\mathcal{E}^+|/|\mathcal{E}|$ represents the ratio of the number of positive edges to the number of all edges. BC-Alpha and BC-OTC [42] are “who-trusts-whom” networks from Bitcoin trading platforms. Wikirfa [43] is a Wikipedia voting network for admin promotion. Slashdot [44] is a signed network from the Slashdot site, where users mark others as friends or foes. Epinions [45] is a review-based trust network from the Epinions platform.

TABLE II
DATA STATISTICS OF SIGNED GRAPHS.

Datasets	$ \mathcal{V} $	$ \mathcal{E} $	$ \mathcal{E}^+ $	$ \mathcal{E}^- $	$\rho(+)$
BC-Alpha ¹	3,783	24,186	22,650	1,536	93.6
BC-OTC ¹	5,881	35,592	32,029	3,563	90.0
Wikirfa ¹	7,634	173,332	135,753	37,579	78.3
Slashdot ¹	82,140	500,481	380,933	119,548	76.1
Epinions ¹	131,580	713,593	589,888	123,705	85.3

¹<https://snap.stanford.edu/data/>

Training and evaluation. Each dataset is randomly split into training (80%), validation (10%), and test (10%) sets. Then, we randomly select labeled edges from the training set according to the specified label ratio ρ_L to form the labeled subset $\mathcal{E}_L$, preserving the positive ratio $\rho(+)$ of edges in the training graph within the selected edges, while the signs of the remaining edges are removed to construct the unlabeled subset $\mathcal{E}_U$. We generate initial node features by applying truncated SVD to the signed adjacency matrix based on $\mathcal{E}_L$. Alternative initialization schemes, such as those based on spectral analysis [46], [47] or tensor factorization [48], [49], can also be adopted. We train each model, including PiGLET, on the training set and select the best hyperparameters based on the validation set. We evaluate link sign prediction performance on the test set. Considering class imbalance, we measure AUC and Macro-F1, which are standard metrics used in prior studies [9]–[11], [19]. We repeat the experiment five times for different random seeds, and report the average with standard deviation.

Baselines. Since the existing SGNNs are not directly applicable to partially observed signed graphs, we heuristically fill in the signs of unlabeled edges and then apply the SGNNs to learn from the resulting graph. The heuristics used for the sign assignment are as follows:

- **Positive assignment (denoted *pos.*):** We simply assign a positive sign to all unlabeled edges in $\mathcal{E}_U$.
- **Predicted assignment (denoted *pred.*):** Initially, we train an SGNN on the set $\mathcal{E}_L$ of labeled edges, and infer hard labels for all unlabeled edges in $\mathcal{E}_U$. We retrain the model and optimizer after reinitialization using the predicted labels.

These heuristics reveal each SGNN’s sensitivity to missing or misspecified signs under a fixed training budget, enabling a direct comparison with our probabilistic soft-label approach. We do not include the heuristic of assigning a negative sign to all unlabeled edges because of its inferior performance.

For SGNNs, we adopt the following representative architectures that are widely used:

- **SGCN** [9]: An extension of GCN that propagates messages separately along balanced and unbalanced paths.
- **SiGAT** [10]: A graph attention network that assigns attention weights to neighbors based on social triad-based attention coefficients.
- **SDGNN** [11]: A signed-directed GNN with four types of aggregators trained via triad-based self-supervision.
- **SNEA** [19]: A neighbor-aware model that combines sign-specific attention with gated aggregation.

TABLE III

PERFORMANCE COMPARISON ON THE SEMI-SUPERVISED LINK SIGN PREDICTION TASK WITH LABEL RATIO $\rho_L = 0.1$. BOLD AND UNDERLINED NUMBERS DENOTE THE BEST AND SECOND-BEST PERFORMANCE PER DATASET, RESPECTIVELY. THE % INC. INDICATES THE PERCENTAGE INCREASE OF PiGLET OVER THE SECOND-BEST BASELINE. PiGLET ACHIEVES THE HIGHEST AUC AND MACRO-F1 SCORES ACROSS ALL TESTED DATASETS.

AUC											
Datasets	SGCN		SiGAT		SDGNN		SNEA		PiGLET (proposed)	% Inc.	
	pos.	pred.	pos.	pred.	pos.	pred.	pos.	pred.			
BC-Alpha	0.748 ± 0.030	0.741 ± 0.039	0.678 ± 0.035	0.654 ± 0.013	0.737 ± 0.022	<u>0.766 ± 0.031</u>	0.712 ± 0.009	0.717 ± 0.026	0.814 ± 0.048	+ 6.3%	
BC-OTC	0.760 ± 0.011	0.746 ± 0.015	0.733 ± 0.024	0.687 ± 0.032	0.776 ± 0.003	<u>0.804 ± 0.015</u>	0.727 ± 0.022	0.710 ± 0.017	0.853 ± 0.011	+ 6.1%	
Wikirfa	0.771 ± 0.003	<u>0.783 ± 0.007</u>	0.674 ± 0.007	0.624 ± 0.006	0.687 ± 0.089	0.772 ± 0.008	0.768 ± 0.005	0.756 ± 0.008	0.842 ± 0.006	+ 7.6%	
Slashdot	0.701 ± 0.005	0.717 ± 0.005	0.675 ± 0.007	0.640 ± 0.029	0.712 ± 0.011	<u>0.761 ± 0.006</u>	0.696 ± 0.002	0.723 ± 0.005	0.799 ± 0.008	+ 4.9%	
Epinions	0.782 ± 0.002	0.830 ± 0.003	0.833 ± 0.010	0.779 ± 0.043	0.699 ± 0.143	<u>0.870 ± 0.009</u>	0.734 ± 0.009	0.782 ± 0.008	0.887 ± 0.003	+ 1.9%	
Macro-F1											
Datasets	SGCN		SiGAT		SDGNN		SNEA		PiGLET (proposed)	% Inc.	
	pos.	pred.	pos.	pred.	pos.	pred.	pos.	pred.			
BC-Alpha	0.633 ± 0.042	0.632 ± 0.010	0.562 ± 0.004	0.544 ± 0.014	0.596 ± 0.021	0.612 ± 0.016	0.616 ± 0.017	<u>0.636 ± 0.025</u>	0.658 ± 0.031	+ 3.3%	
BC-OTC	0.654 ± 0.009	0.657 ± 0.013	0.588 ± 0.026	0.556 ± 0.029	<u>0.658 ± 0.013</u>	0.654 ± 0.022	0.645 ± 0.006	0.642 ± 0.008	0.731 ± 0.017	+ 11.0%	
Wikirfa	0.663 ± 0.006	0.658 ± 0.008	0.586 ± 0.011	0.553 ± 0.005	0.572 ± 0.069	0.654 ± 0.008	0.681 ± 0.001	0.671 ± 0.009	0.729 ± 0.006	+ 7.0%	
Slashdot	0.622 ± 0.007	0.622 ± 0.004	0.615 ± 0.011	0.559 ± 0.035	0.598 ± 0.010	<u>0.649 ± 0.019</u>	0.635 ± 0.003	0.635 ± 0.005	0.693 ± 0.015	+ 6.8%	
Epinions	0.687 ± 0.009	0.684 ± 0.012	<u>0.731 ± 0.012</u>	0.618 ± 0.039	0.587 ± 0.070	0.705 ± 0.007	0.718 ± 0.002	0.712 ± 0.005	0.801 ± 0.009	+ 9.6%	

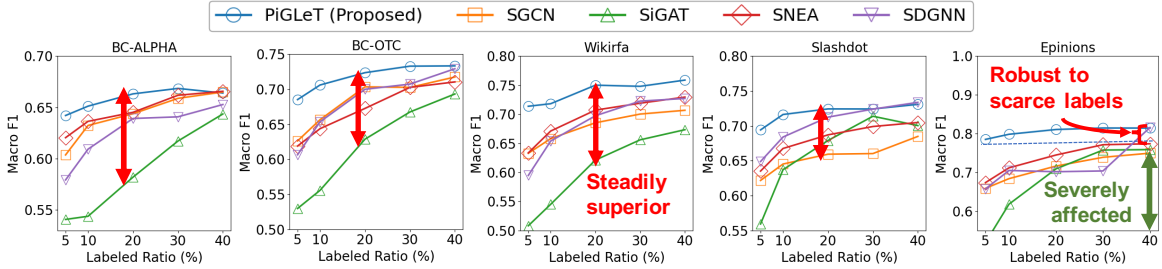


Fig. 3. Performance comparison across different label ratios ρ_L in terms of Macro-F1 score. We evaluate the performance of PiGLET and baseline methods by varying ρ_L from 0.05 to 0.4. PiGLET achieves higher accuracy than the baselines across most label ratios, exhibiting stable and superior performance even at lower label ratios. These results show that PiGLET effectively leverages limited labeled supervision while mitigating uncertainty from unlabeled edges.

Hyperparameter tuning. We grid search over hyperparameters for all models, including PiGLET, to ensure a fair comparison. For the baselines, we search for hyperparameters suggested in their respective papers. For PiGLET, we search over the ranges $\{1, 2, 3, 4, 5, 6\}$ for λ_S and $\{0.1, 0.5, 1, 5, 10\}$ for λ_U . Following the standard settings from SGNN, we fix the number L of layers to 2, the number T of epochs to 100, the embedding dimension d to 64, the learning rate η to 0.01, and the weight decay to 10^{-5} for all methods.

Machine and implementation. We use a workstation with GTX 2080 Ti. Our method PiGLET was implemented using Pytorch 1.13.1 in Python 3.8.20. For the baseline SGNNs, we use the open-source implementation provided by [41].

B. Performance Comparison (Q1)

We compare PiGLET with the baselines on the link sign prediction task, where the label ratio ρ_L is set to 0.1 (i.e., only 10% of the edges in the training set are labeled). Table III shows a clear and consistent superiority of PiGLET over all baselines. Specifically, PiGLET achieves the highest AUC and Macro-F1 scores across all evaluated datasets, delivering improvements of up to 7.6% in AUC and 11.0% in Macro-F1. We further analyze the performance of PiGLET by varying ρ_L from 0.05 to 0.4, comparing it against the baselines using the predicted assignment. Fig. 3 shows that PiGLET provides

robust performance for all tested label ratios, while the baseline methods consistently show lower performance and tend to become unstable as ρ_L decreases. These results indicate that the naïve sign assignments for Problem 1 introduce noise and increase uncertainty in the SGNNs due to unreliable predictions, whereas PiGLET effectively addresses them through iterative refinement and confidence-based propagation.

C. Ablation Study (Q2)

We analyze the contribution of each core component in PiGLET by evaluating the following ablated variants:

- **PiGLET-one** softly propagates only through the sign with higher predicted probability.
- **PiGLET-hard** assigns each edge the sign with the higher predicted probability and treats it as labeled.
- **PiGLET-w/o-conf** removes the confidence weighting.
- **PiGLET-w/o-attn** retains the confidence weighting but disables the attention-based propagation.
- **PiGLET-w/o-conf&attn** removes both confidence and attention, performing uniform aggregation.

Table IV presents the performance of PiGLET and its ablated variants; PiGLET consistently outperforms the others. Propagating along only one side, whether softly (PiGLET-one) or deterministically (PiGLET-hard), degrades performance, underscoring the advantage of inclusive two-way

TABLE IV

ABLATION STUDY. PiGLET OUTPERFORMS ITS ABLATED VARIANTS, SHOWING THE EFFECTIVENESS OF TWO-WAY AND SOFT-LABEL BASED PROPAGATION, ATTENTION, AND CONFIDENCE-WEIGHTING MECHANISMS.

Methods	BC-Alpha		BC-OTC		Wikirfa		Slashdot		Epinions	
	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1
one	0.748	0.621	0.791	0.660	0.805	0.686	0.740	0.660	0.821	0.753
hard	0.768	0.599	0.793	0.641	0.753	0.657	0.720	0.622	0.819	0.742
w/o-conf	0.784	0.652	0.822	0.704	0.839	0.724	0.782	0.679	0.884	0.811
w/o-attn	0.744	0.623	0.787	0.651	0.823	0.688	0.681	0.623	0.831	0.718
w/o-conf&attn	0.713	0.622	0.785	0.665	0.817	0.687	0.681	0.594	0.832	0.706
PiGLET	0.814	0.658	0.853	0.731	0.842	0.729	0.799	0.693	0.887	0.801

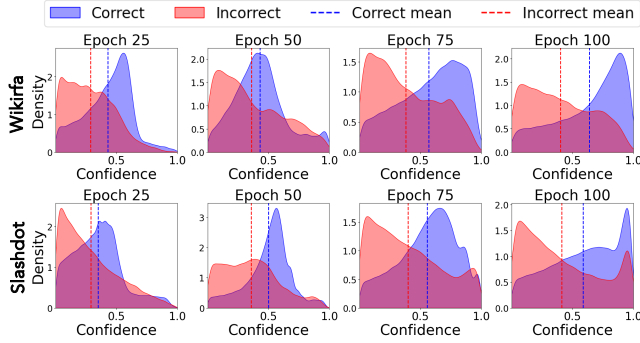


Fig. 4. Confidence distribution of unlabeled edges over training epochs. Blue and red distributions represent correctly and incorrectly predicted unlabeled edges, respectively, where dashed lines denote the mean of each distribution. As training progresses, the correctly predicted edges exhibit increased confidence, whereas incorrectly predicted edges maintain dispersed confidence distributions, reflecting persistent model uncertainty.

message passing. Removing either confidence weighting or relation-aware attention causes a drop, while removing both components leads to a substantial decline, especially on BC-Alpha and Slashdot. These results indicate that each component effectively addresses uncertainty in learning from pseudo-labeled signed graphs, and their synergy significantly improves performance on semi-supervised link sign prediction.

D. Confidence Analysis (Q3)

We analyze the confidence distribution of predictions over training epochs to better understand how PiGLET handles uncertainty from unlabeled edges. Specifically, we measure the confidence of each unlabeled edge at each epoch, and categorize them based on whether their predicted labels match the ground truth (i.e., correct or incorrect). Fig. 4 shows that the confidence distributions for correctly (blue) and incorrectly (red) predicted unlabeled edges over training epochs. As training progresses, the distribution of correct predictions becomes right-skewed, while that of incorrect predictions remains low and dispersed. This suggests that PiGLET progressively improves in distinguishing between reliable and unreliable predictions. Note that the low confidence for most incorrect edges implies that these samples are inherently difficult to classify, possibly due to limited supervision or noisy signals. Thus, PiGLET gradually focuses on aggregating reliable messages from high-confidence edges, while diminishing the impact of uncertain messages from low-confidence ones. This confidence-based aggregation enhances the quality of node representations and link sign prediction accuracy.

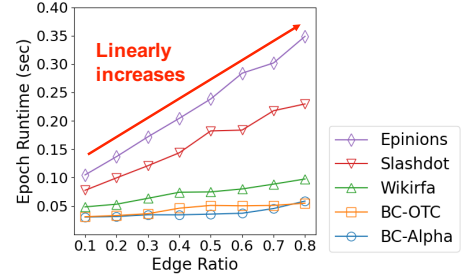


Fig. 5. Scalability analysis of PiGLET with respect to the edge ratio on various datasets. The runtime per epoch increases linearly with the edge ratio, particularly for larger datasets such as Epinions and Slashdot.

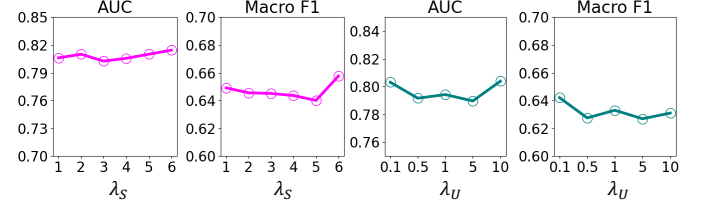


Fig. 6. Hyperparameter sensitivity analysis of PiGLET on BC-Alpha dataset. We examine the impact of varying structural loss weight λ_S (pink) and pseudo-label loss weight λ_U (green) on AUC and Macro-F1. Both metrics demonstrate stable behavior across a wide range of values, indicating robustness and ease of hyperparameter tuning for PiGLET.

E. Scalability (Q4)

We analyze the scalability of PiGLET by randomly sampling edges from each dataset with ratios from 0.1 to 0.8, and measuring the average runtime per epoch during training. As shown in Fig. 5, PiGLET scales linearly with the edge ratio (i.e., the number of edges) across all tested datasets, with this trend being especially evident in larger datasets such as Slashdot and Epinions. These empirical results are consistent with Theorem 2, confirming that PiGLET is scalable with respect to the number of edges.

F. Effect of Hyperparameters (Q5)

We examine the effect of λ_S and λ_U , the main hyperparameters of PiGLET on the BC-Alpha dataset. Fig. 6 illustrates the impact of varying these hyperparameters on AUC and Macro-F1. Specifically, changes in λ_S (pink) and λ_U (green) produce only slight fluctuations in performance, suggesting that optimal results can be easily achieved without extensive hyperparameter tuning.

V. CONCLUSION

In this paper, we propose PiGLET, a novel method for accurate link sign prediction on partially observed signed graphs. We newly formulate the problem of semi-supervised link sign prediction. To address this problem, we propose (1) probabilistic pseudo-labeling for unlabeled edges, (2) confidence-weighted relation-aware attention via balance theory-based message aggregation, and (3) iterative refinement that gradually improves both embeddings and predictions. We theoretically justify the iterative refinement mechanism and show that it scales linearly. Extensive experiments show that PiGLET consistently outperforms baselines on the target task, achieving improvement in both AUC and Macro-F1.

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